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(54) **NAMESPACES FOR PROTECTED MULTI-TENANT PACKET PROCESSING PIPELINES**

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(57) **ABSTRACT**

A programmable multi-stage packet processing pipeline executing in a data plane of a software-defined networking (SDN) network handles data packets according to objects in a directed graph that are programmed using multiple separate namespaces. Each namespace provides isolation for programming in the processing stages to provide protection for the pipeline that may span across multiple different tenant domains. While programming is handled using separate namespaces, the programmed stages are converted into executable code that runs in a single data plane of the SDN network to optimize data packet flow while minimizing latency.

1000

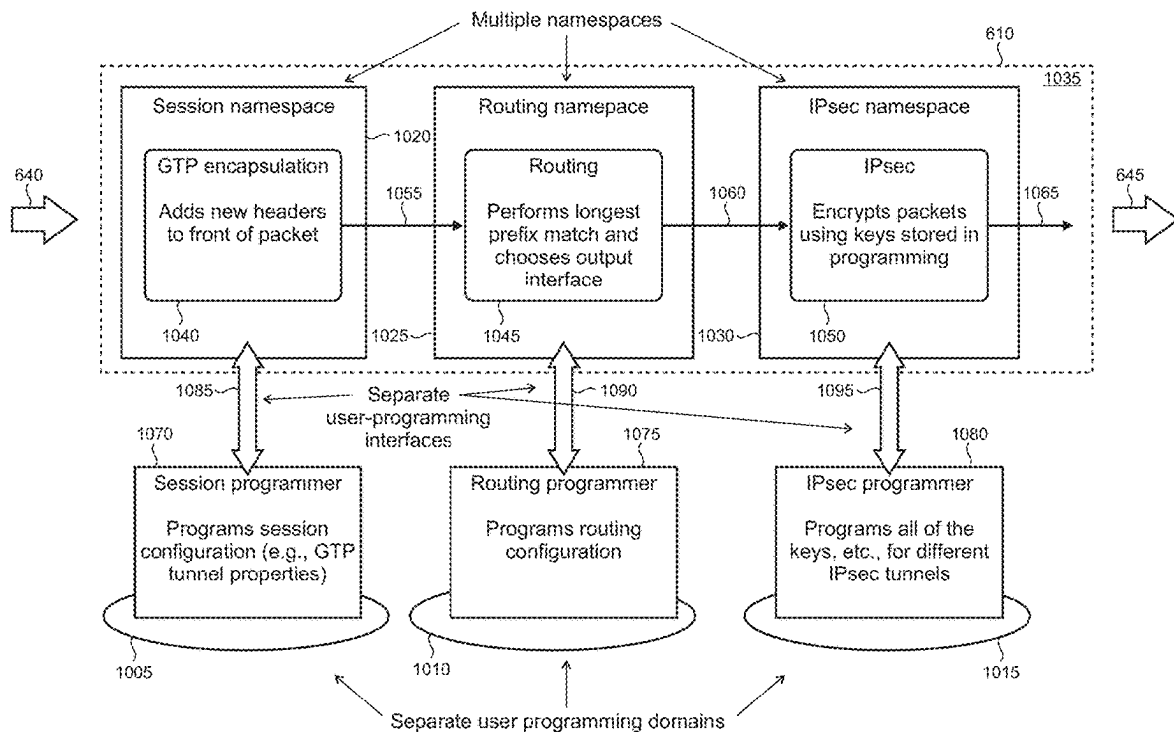


FIG 1

100

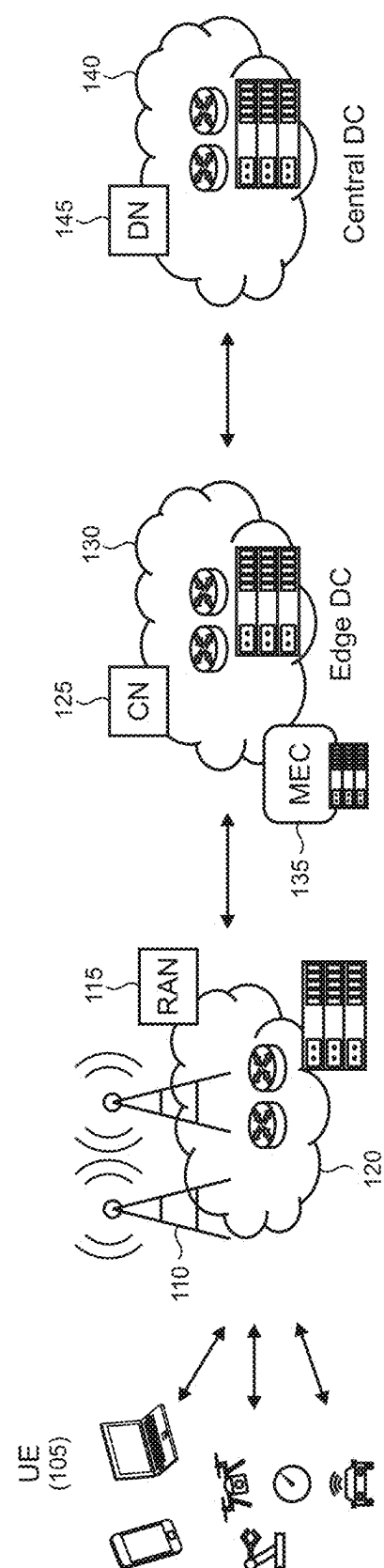
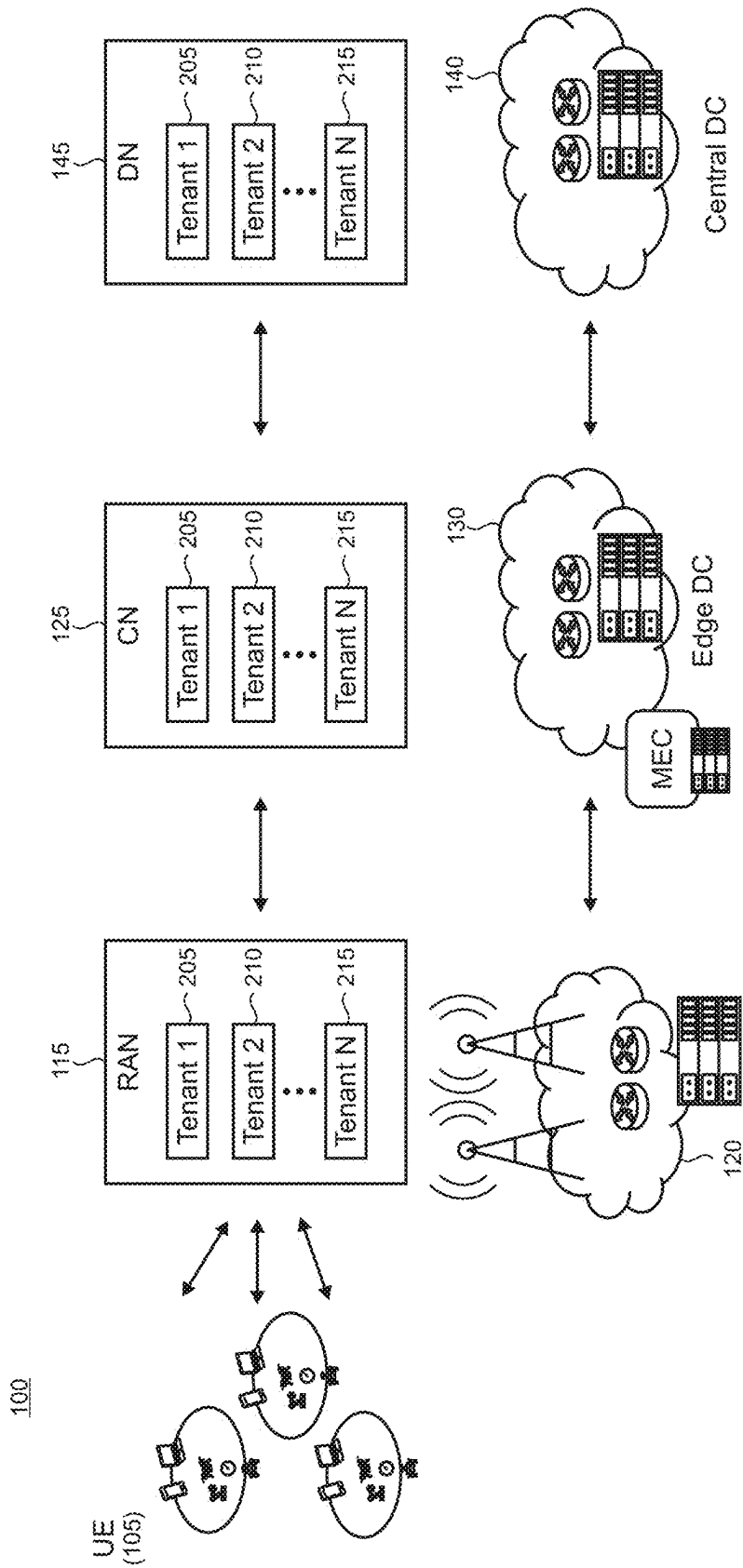


FIG 2



UE = User equipment
RAN = Radio access network
CN = Core network
DN = Data network
MEC = Multi-access edge compute

FIG 3
300

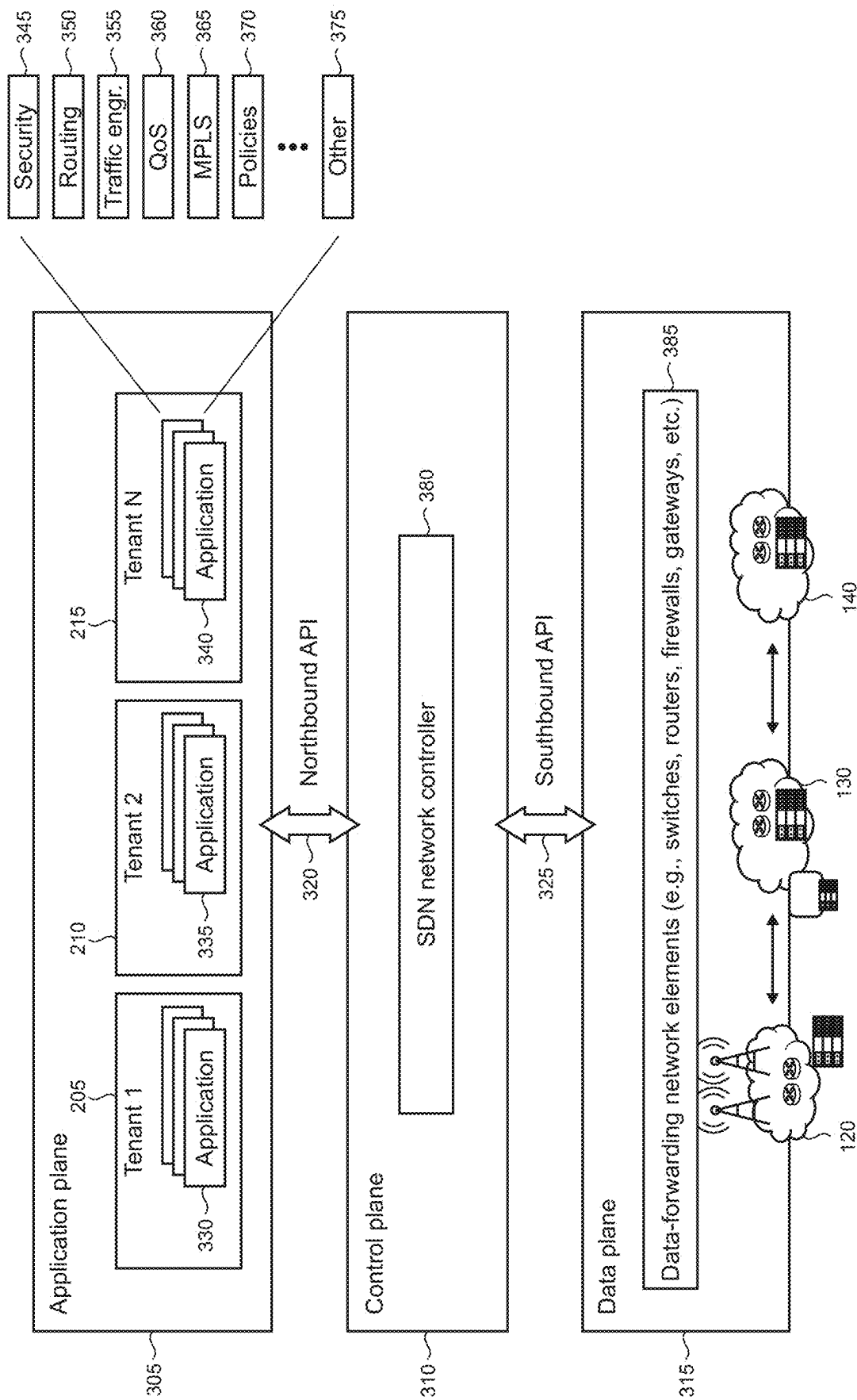


FIG 4
400

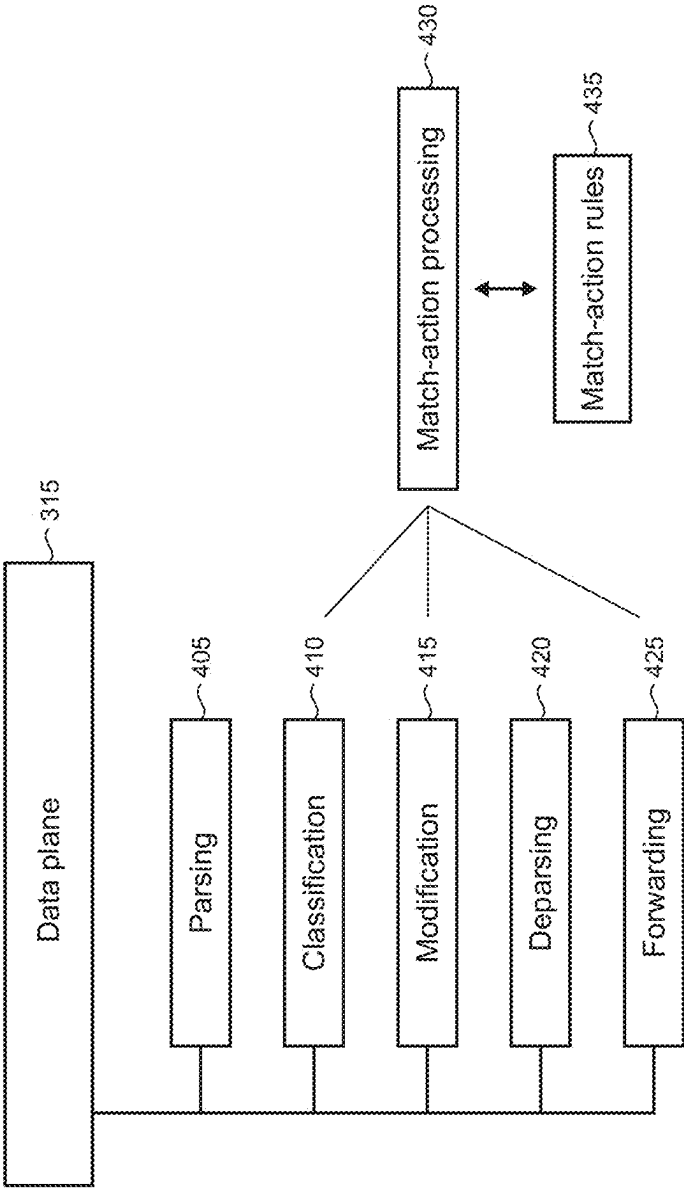


FIG 5

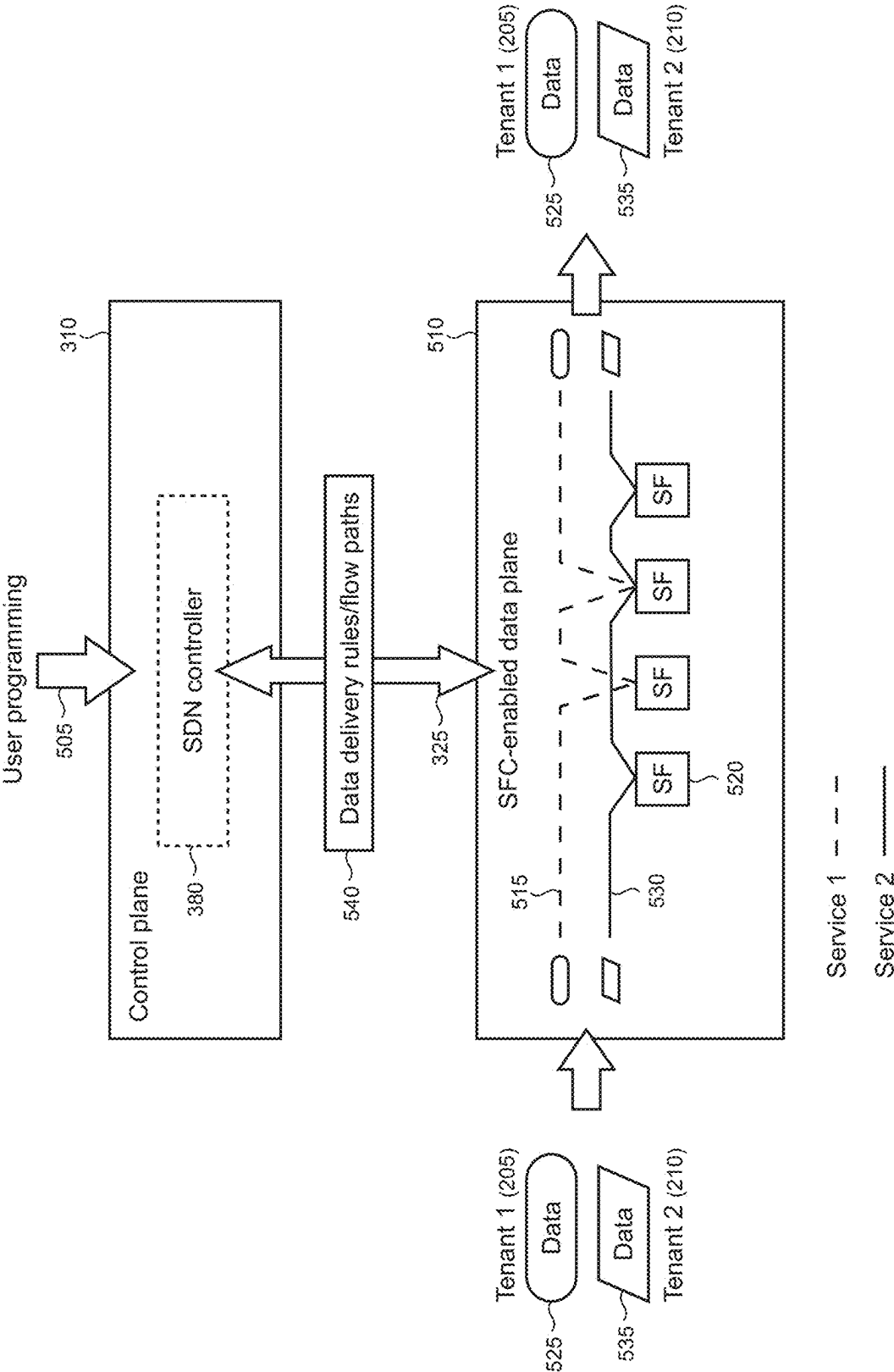


FIG 6

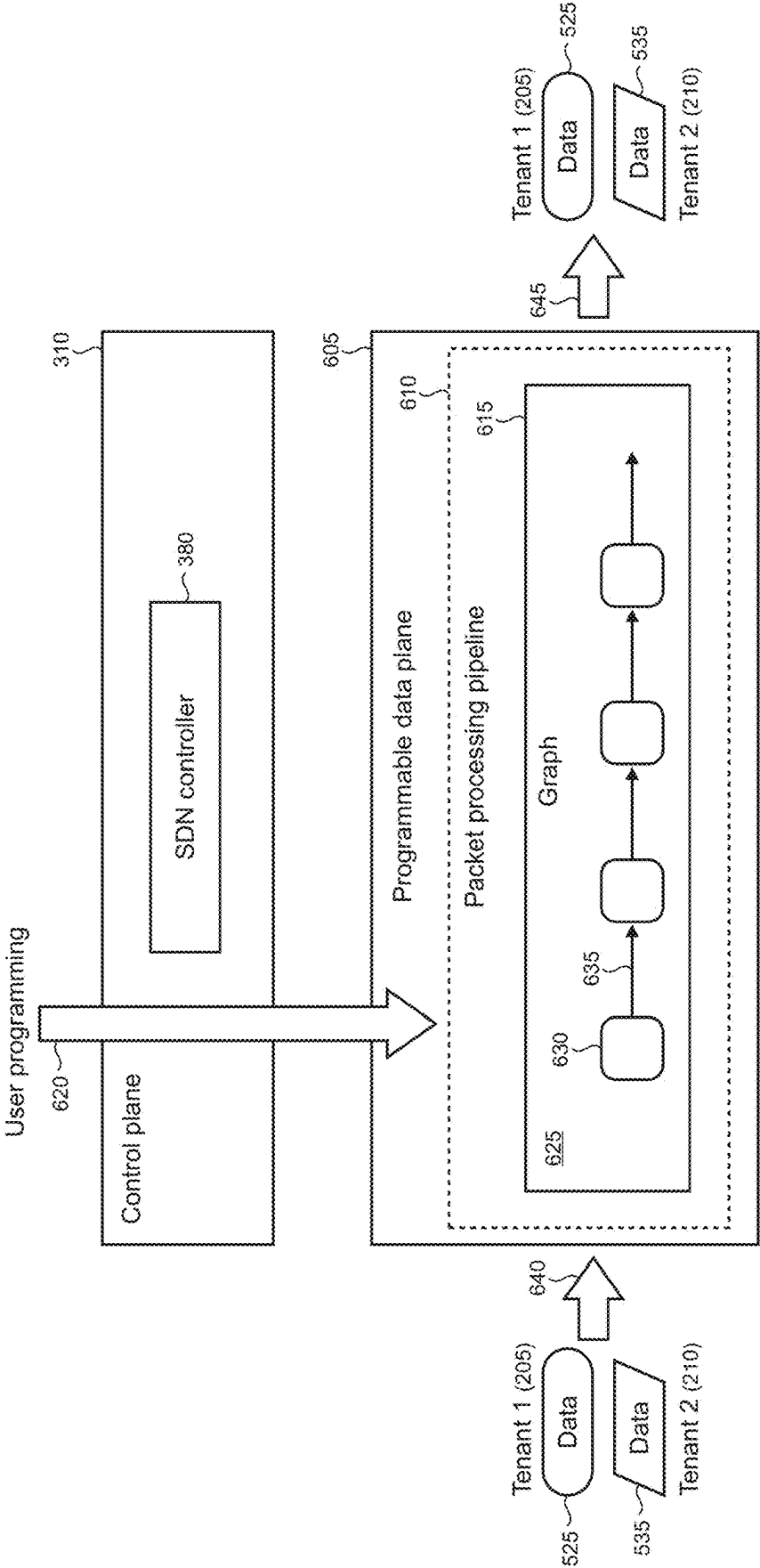


FIG 7

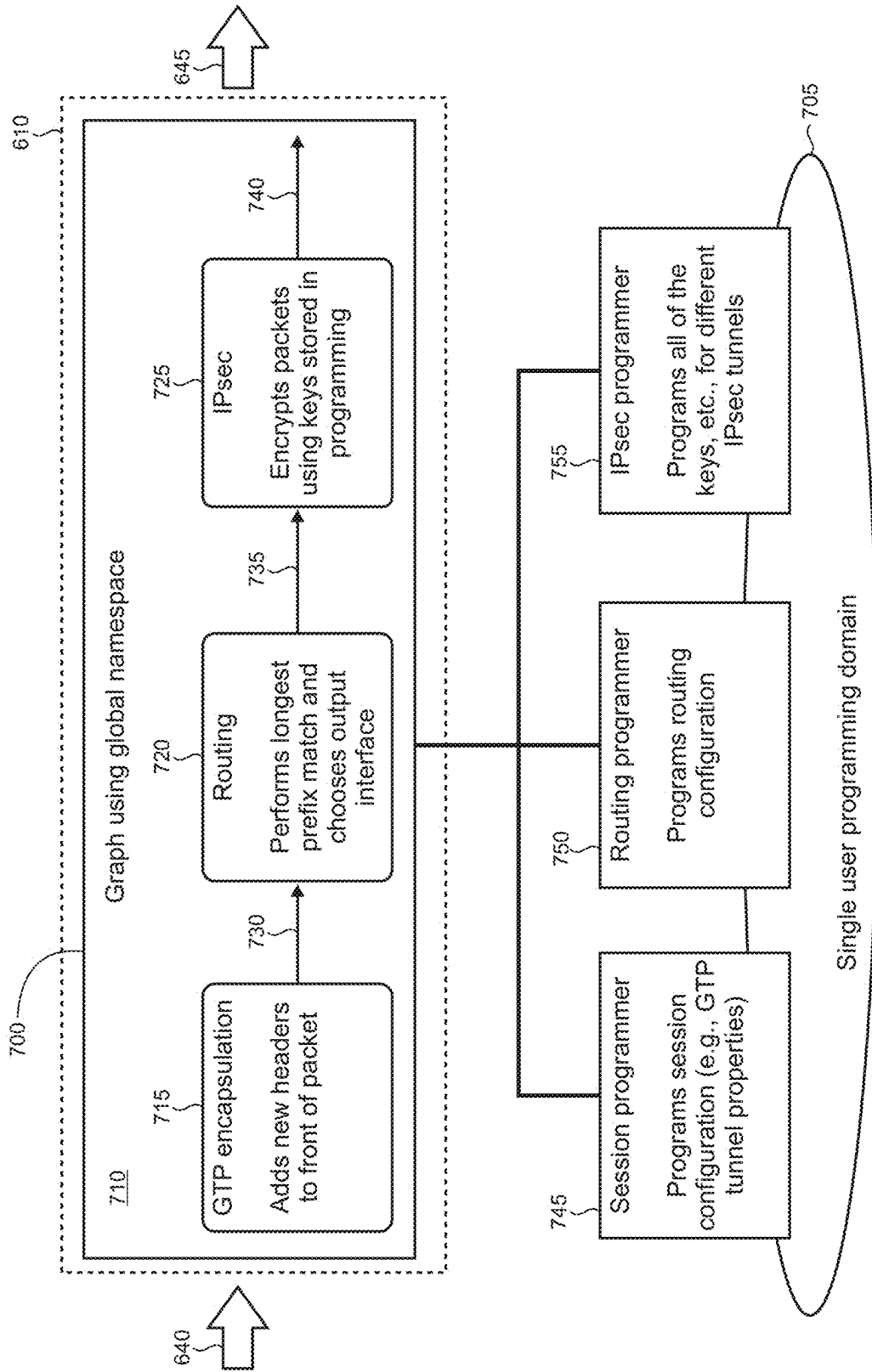


FIG 8

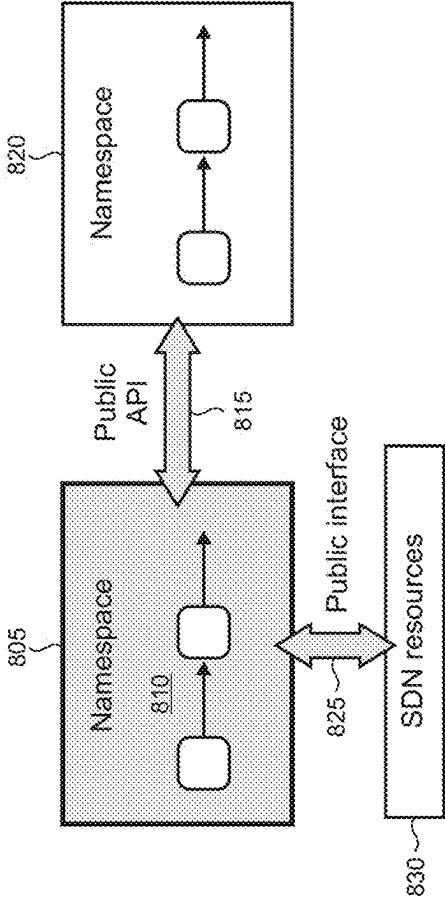


FIG 9A

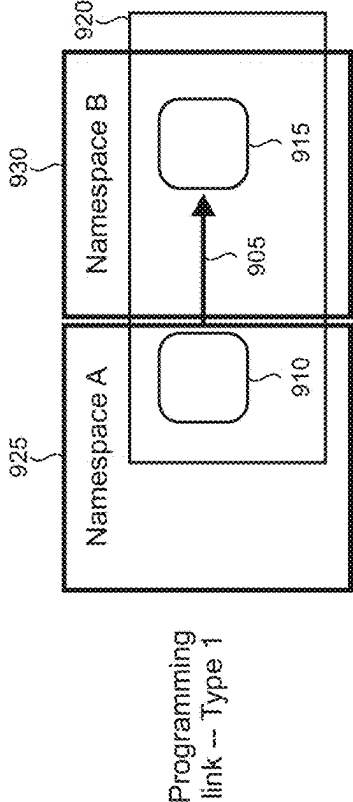


FIG 9B

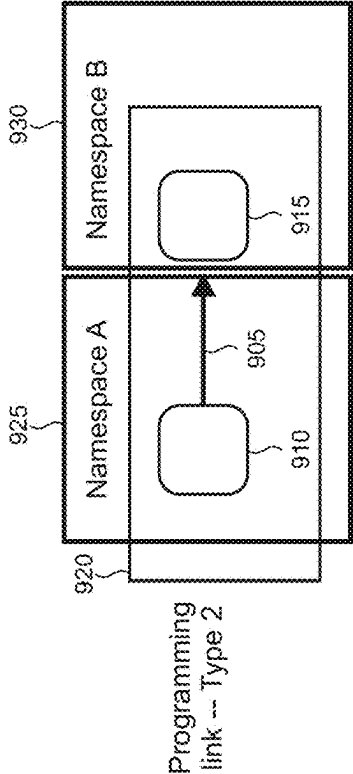
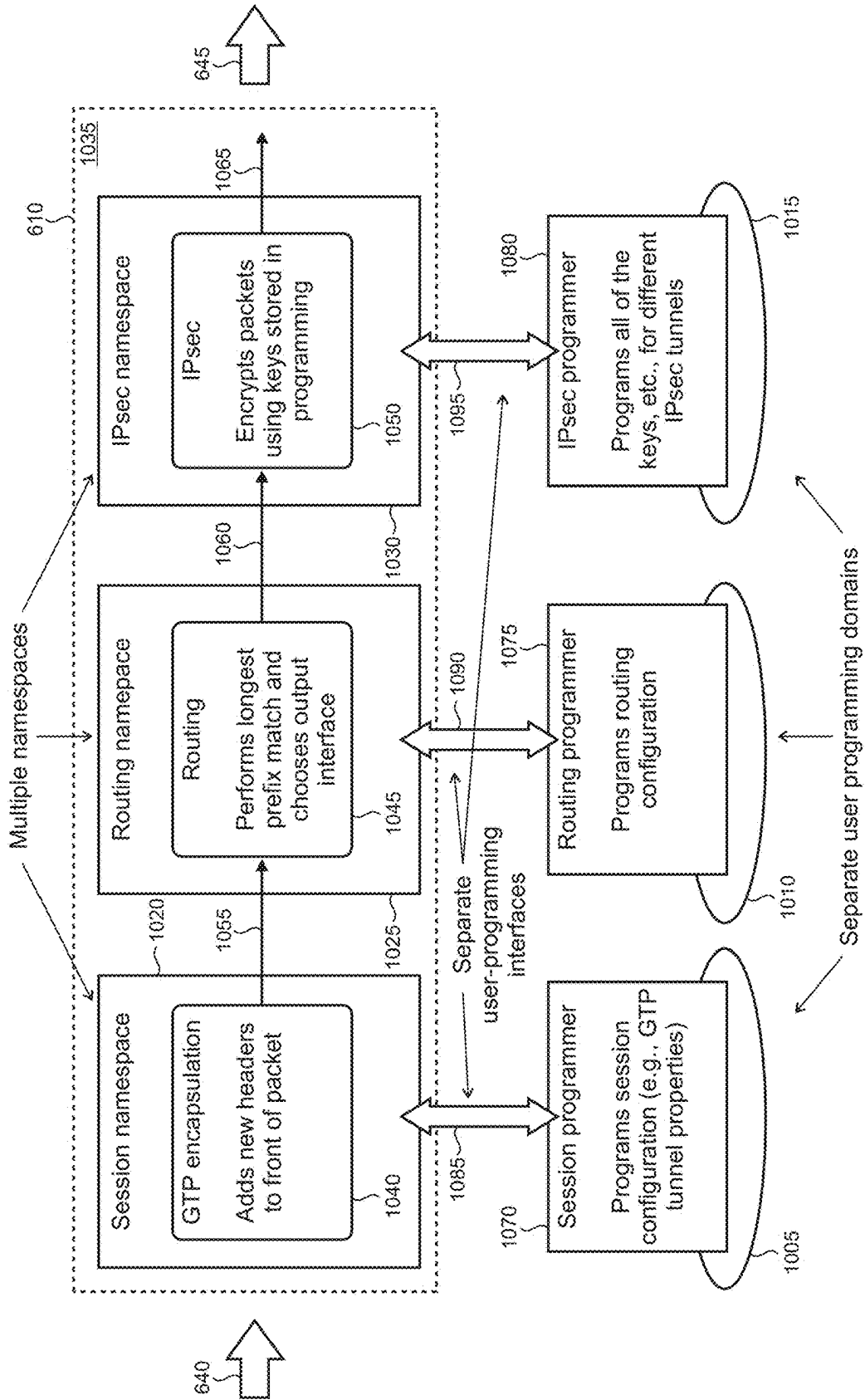


FIG 10
1000



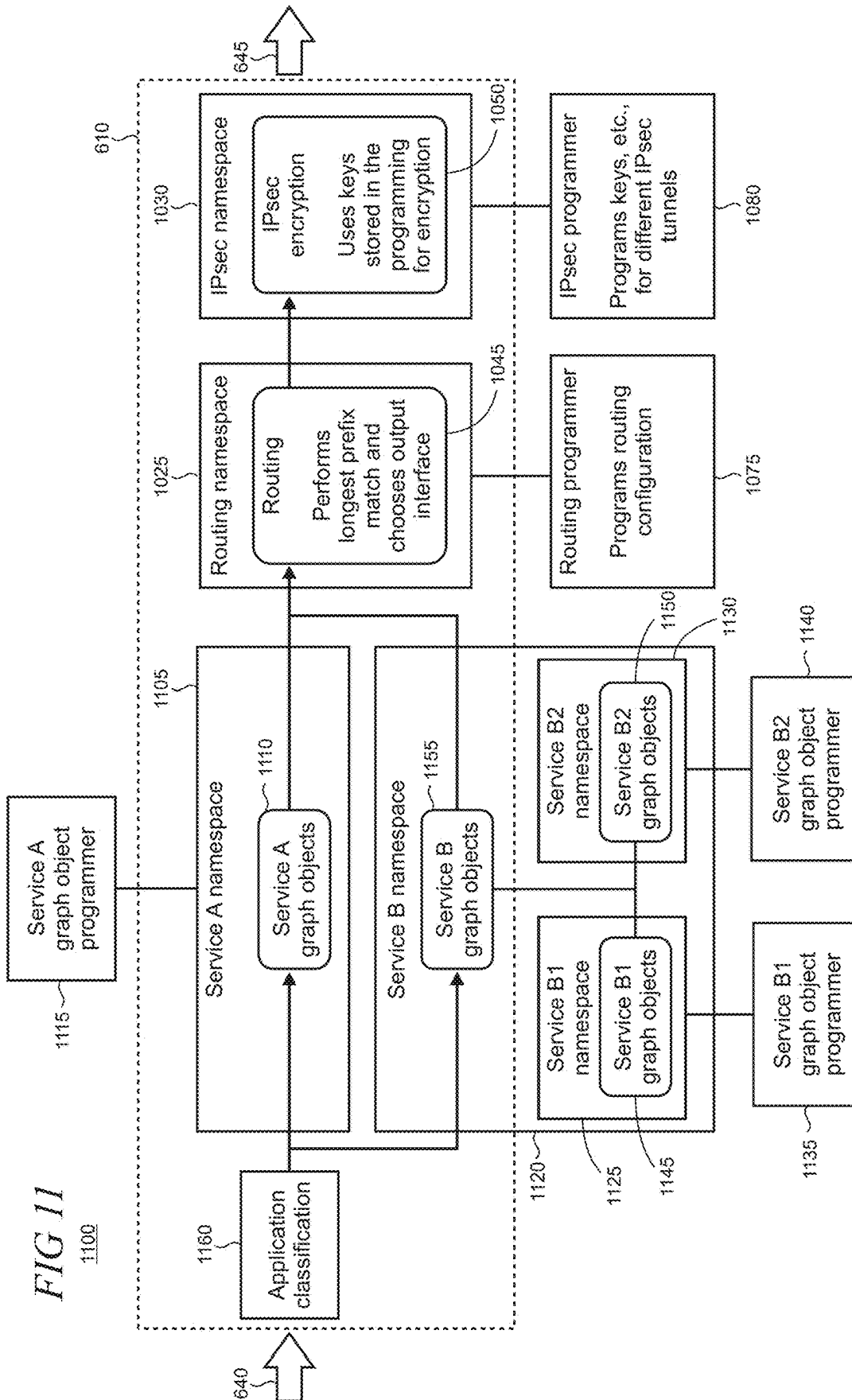


FIG 12

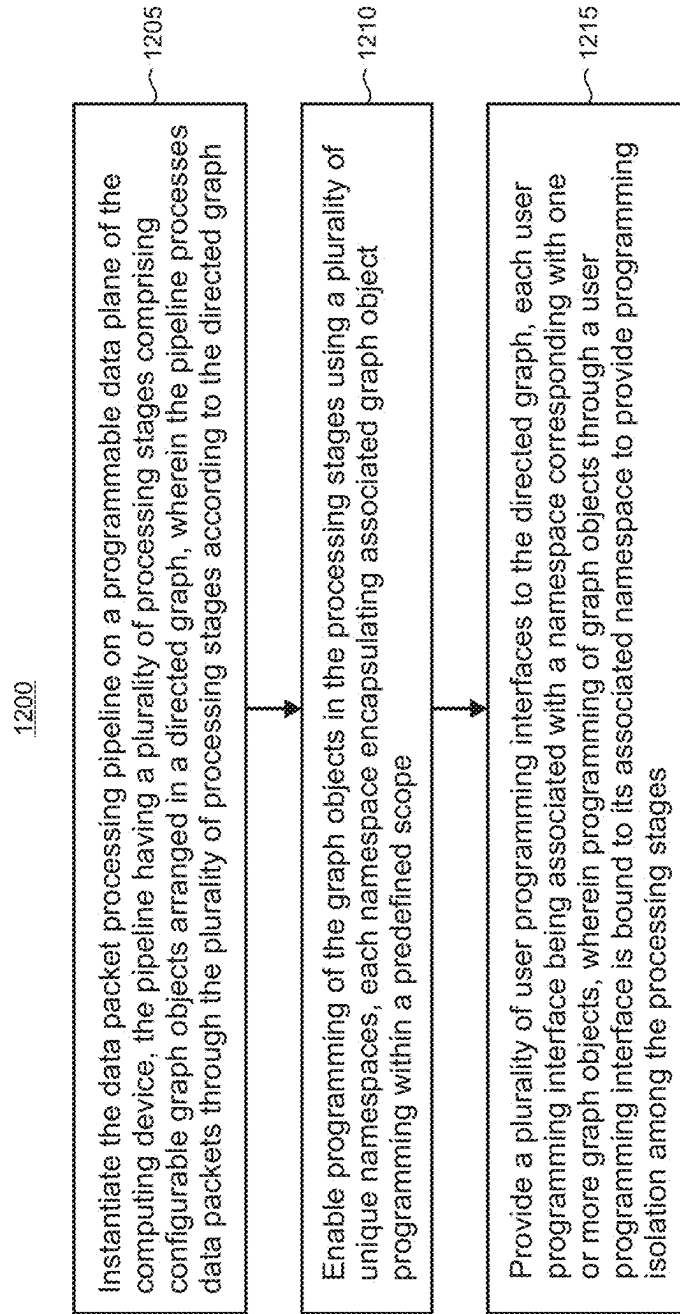


FIG 13

1300

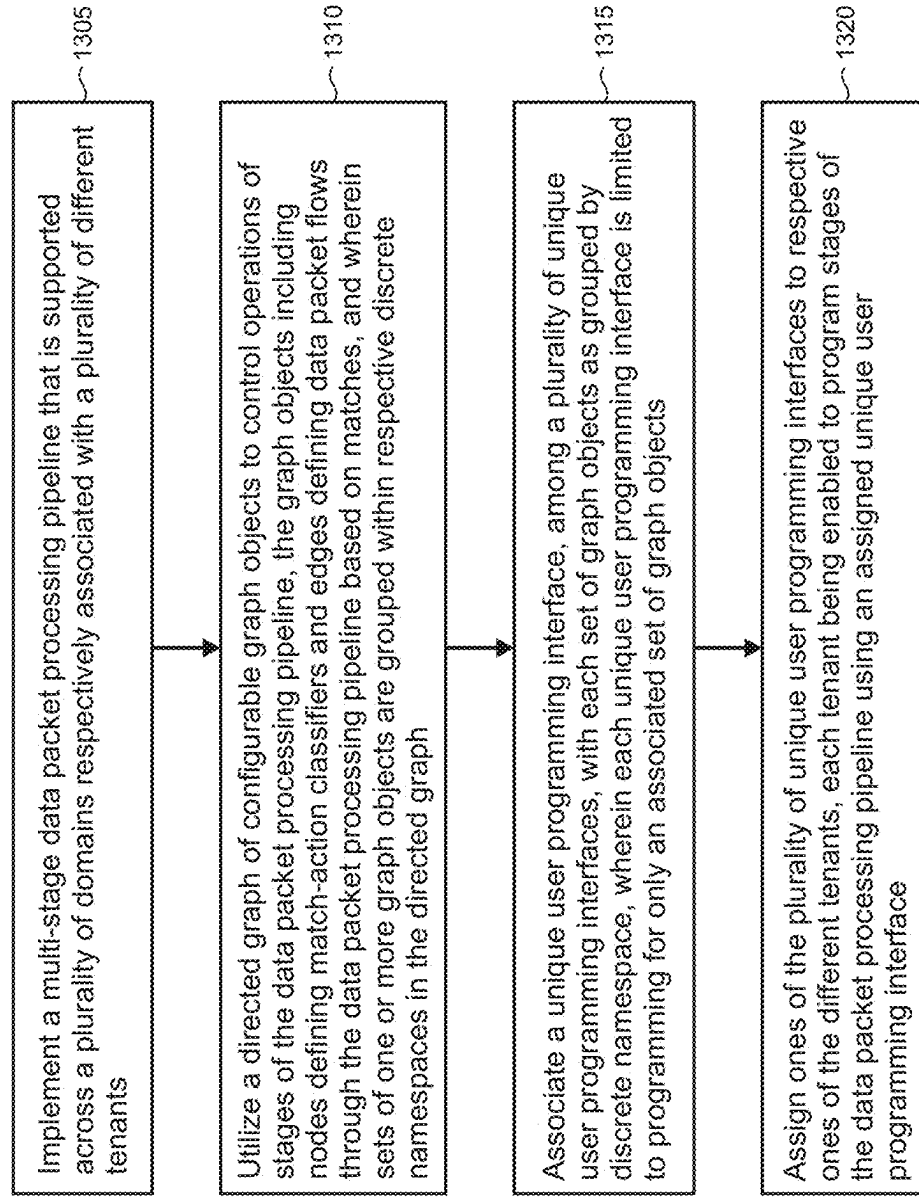


FIG 14

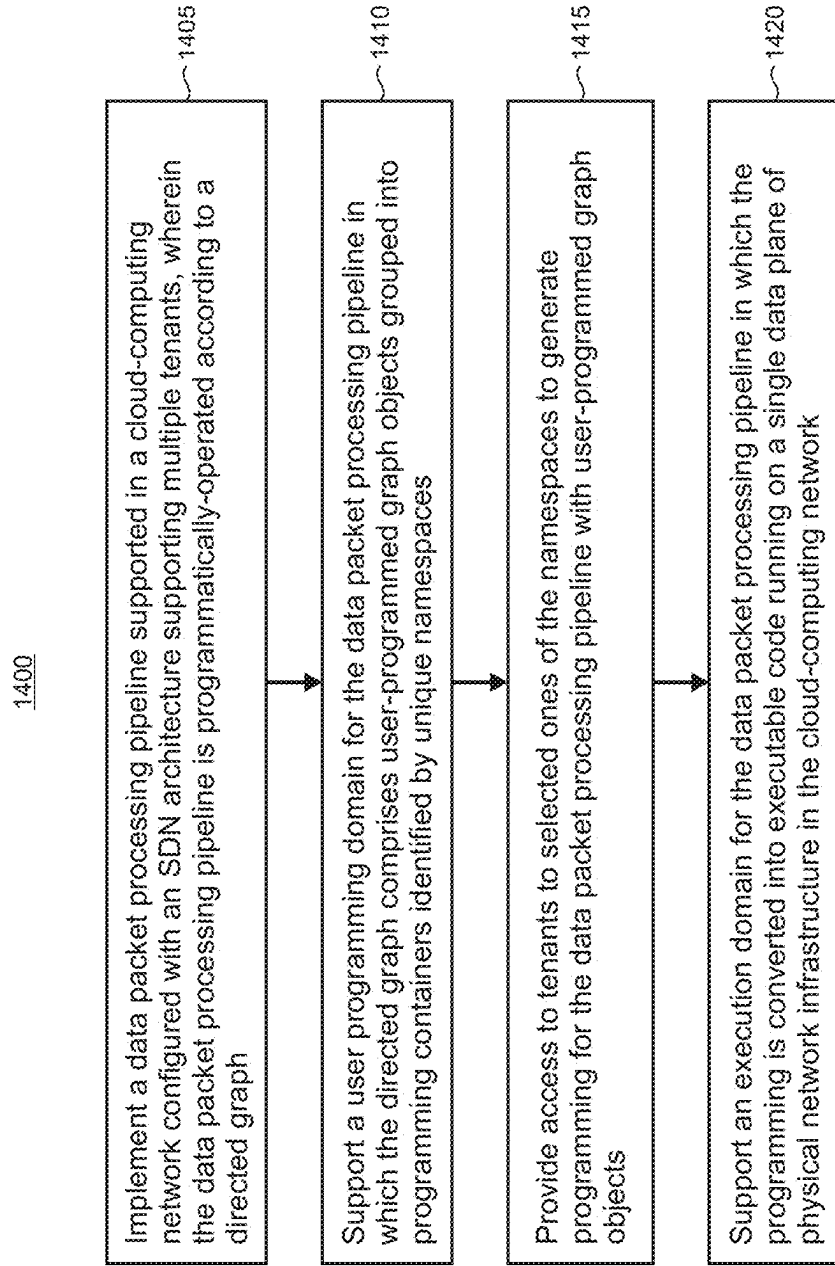


FIG 15
1500

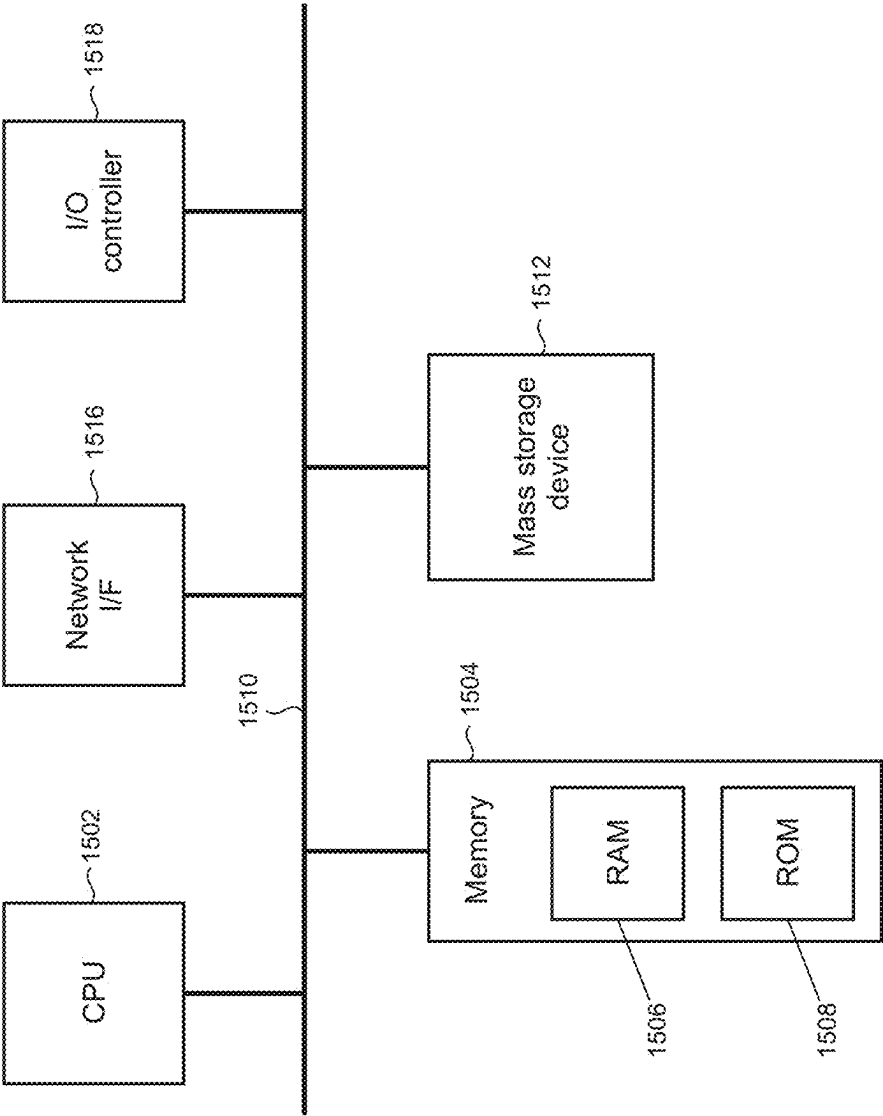


FIG 16

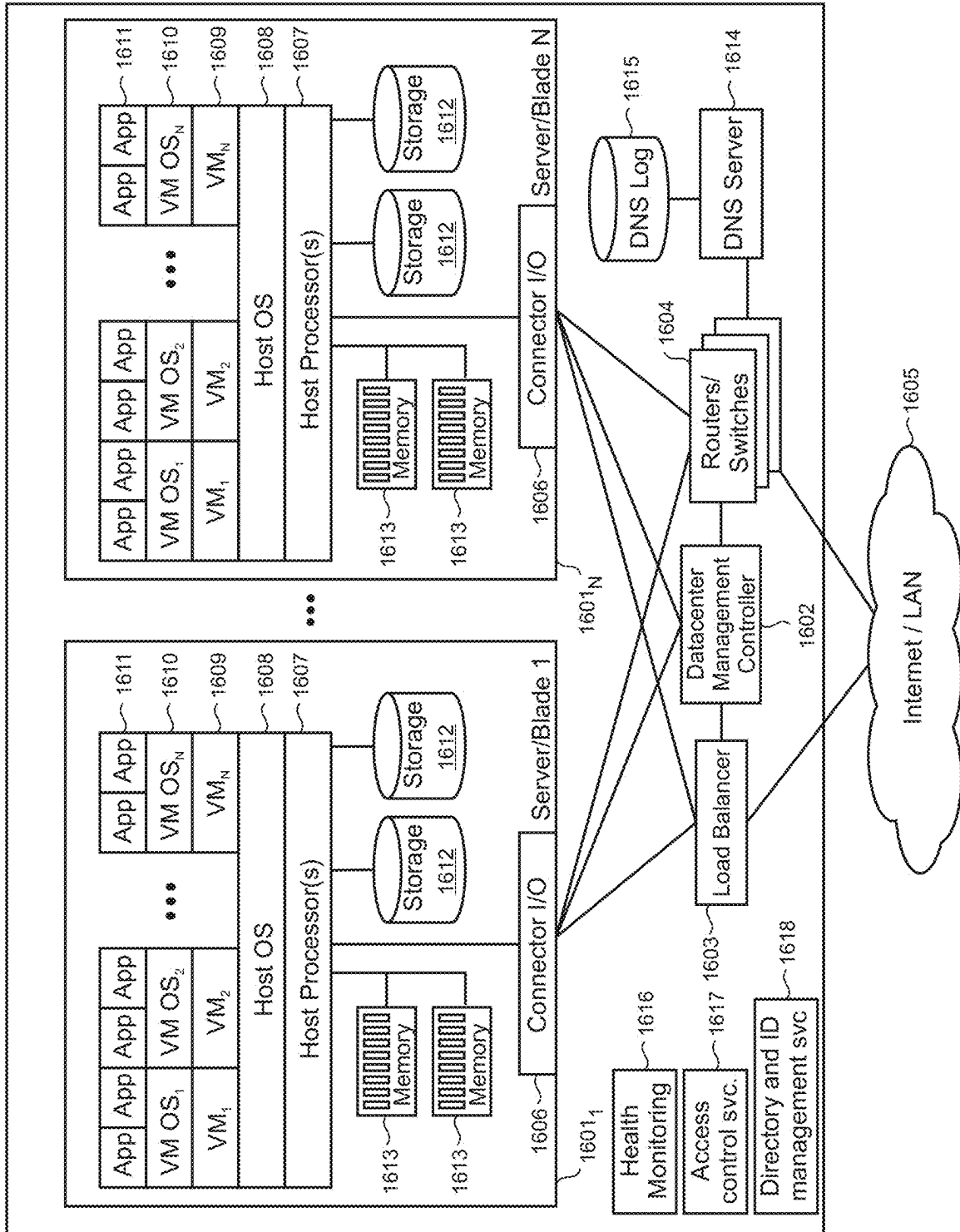
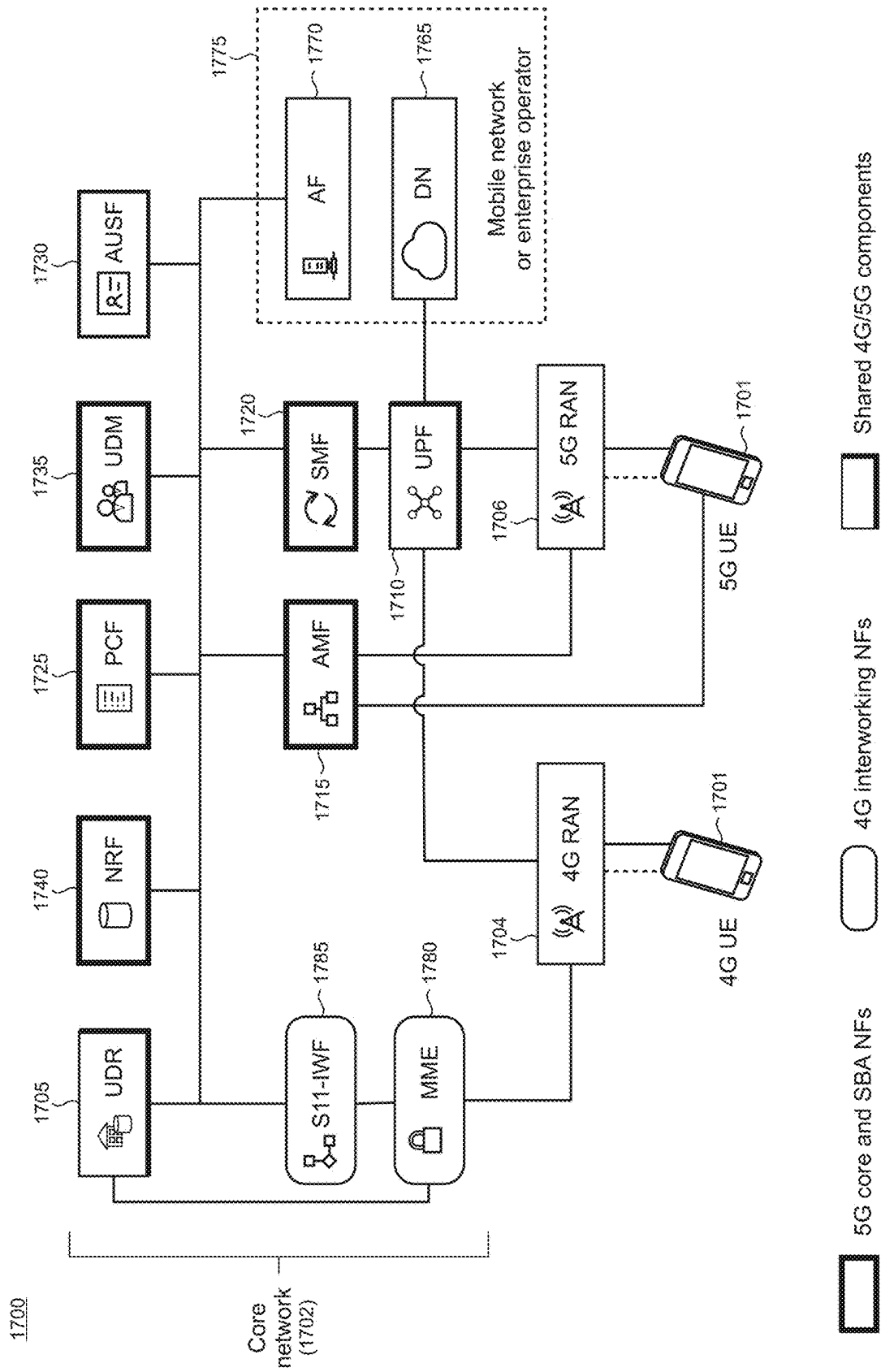


FIG 17



NAMESPACES FOR PROTECTED MULTI-TENANT PACKET PROCESSING PIPELINES

BACKGROUND

[0001] Information technology (IT) resources, such as compute and storage resources, are increasingly distributed into cloud environments that share the underlying IT infrastructure among multiple tenants. Data processing in either or both on-premises and cloud infrastructure may be handled by a pipeline that processes data packets through various stages.

SUMMARY

[0002] Disclosed are a method, computer-readable storage device, and computing device configured for processing data packets in a multi-stage pipeline in a communications network that is programmable according to a directed graph of objects representing processing stages and packet flows through the pipeline. Rather than utilize a single global namespace for programming the entire directed graph, multiple unique namespaces function as containers that are associated with groups of graph objects. Users create and define graph objects through respective programming interfaces to the namespaces. Each unique namespace encapsulates the associated graph object programming within a predefined scope and defines public interfaces for other namespaces to access, and defines interfaces to network resources. Isolation is enforced among groups of graph objects, which are only programmable by users through assigned programming interfaces to associated namespaces, to provide protection to the pipeline against security vulnerabilities by limiting users' programming access to predetermined parts of the pipeline that they are authorized to access.

[0003] Users work through assigned interfaces to chain groups of graph objects together to form services in a programming domain. However, the separation of namespaces and programming interfaces is only enforced in the programming domain—chained services are instantiated at runtime in an execution domain by mapping all of the namespace programming into a single graph. The graph objects are converted into code that executes in a data plane of a software-defined networking (SDN) network. Such mapping ensures that the data plane operates efficiently with minimal latency by reducing the processing overhead associated with chaining services conventionally (e.g., using packet queuing when passing packets among service functions).

[0004] In an illustrative embodiment, the programmable graph objects include nodes and edges in which the nodes represent match-action classifiers and the edges represent matches that determine packet flow between processing stages. The programming of the graph objects is bound to its associated namespace. The match-action classifiers use match-action rules to process packets, for example, by matching filtering criteria to fields in packet headers and responsively applying actions to the packets associated with the matched rules such as discarding, modifying, or forwarding the packet to other nodes.

[0005] In an illustrative embodiment, a packet processing pipeline spans multiple tenant domains instantiated in an SDN network supporting the pipeline. Programming inter-

faces to namespaces and the associated graph objects that control pipeline processing are distributed and assigned among the different tenants. Cross-tenant programming is enabled through application programming interfaces that are defined and exposed by the namespaces while maintaining the pipeline protection and security benefits of the separate isolated programming domains.

[0006] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure. It will be appreciated that the above-described subject matter may be implemented as a computer-controlled apparatus, a computer process, a computing system, or as an article of manufacture such as one or more computer-readable storage media. These and various other features will be apparent from a reading of the following Detailed Description and a review of the associated drawings.

DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows an illustrative 5G (5th generation) mobile network with which principles associated with the present namespaces for protected multi-tenant packet processing pipelines are applicable;

[0008] FIG. 2 shows an illustrative multi-tenant use case in which different tenants commonly share 5G network infrastructure and resources;

[0009] FIG. 3 shows an illustrative software-defined networking (SDN) architecture that provides for an application plane that is supported by separate control and data planes;

[0010] FIG. 4 shows illustrative data plane functions;

[0011] FIG. 5 shows an illustrative example of multi-tenant service function chaining using an SDN controller configured to support user programming;

[0012] FIG. 6 shows a programmable data plane in which a protected multi-tenant packet processing pipeline is operated in accordance with a directed graph;

[0013] FIG. 7 shows an illustrative directed graph using a global namespace that is programmed through a single user programming domain;

[0014] FIG. 8 shows an illustrative namespace that encapsulates graph objects and defines interfaces to other namespaces and SDN resources;

[0015] FIGS. 9A and 9B show illustrative namespaces that may be used to program edge graph objects;

[0016] FIG. 10 shows an illustrative directed graph that is programmed through separate and individual user programming domains using multiple namespaces;

[0017] FIG. 11 shows an illustrative directed graph that is composed of services programmed using a layered approach in which some namespaces are hidden;

[0018] FIGS. 12, 13, and 14 show illustrative methods that may be performed when implementing the present namespaces for protected multi-tenant packet processing pipelines;

[0019] FIG. 15 is a block diagram of an illustrative server or computing device that may be used at least in part to implement the present namespaces for protected multi-tenant packet processing pipelines;

[0020] FIG. 16 is a block diagram of an illustrative data-center that may be used at least in part to implement the present namespaces for protected multi-tenant packet processing pipelines; and

[0021] FIG. 17 shows details of an illustrative 4G/5G core network implemented using an SDN architecture.

[0022] Like reference numerals indicate like elements in the drawings. Elements are not drawn to scale unless otherwise indicated.

DETAILED DESCRIPTION

[0023] The development of service-based network architectures in the fifth generation (5G) of mobile networks has provided technologies including Network Function Virtualization (NFV) and Software-defined Networking (SDN) that enable a wide variety of services to be flexibly and dynamically provisioned. An additional feature of 5G service-based architectures includes the concept of separating control and data plane functionalities to facilitate independent scalability, evolution, and flexible deployment of services using cloud-computing, implemented across, for example, centrally-located computing infrastructure and/or distributed computing infrastructure.

[0024] Wireless 5G networking is frequently utilized in communication networks that can encompass consumer and enterprise networks to support access to a variety of applications, services, and resources including the Internet. Applications and services commonly leverage the high bandwidth and low latency enabled by modern communication networks to provide compelling user experiences and support feature-rich capabilities for a wide variety of connected devices (both fixed and mobile).

[0025] A common approach to service-based network data processing utilizes a pipeline design in which packets of data having a defined format in workloads from various and different network sources are passed among pipeline stages. Packet processing pipelines are configured to perform various operations on packets such as applying policies, modifying or transforming data, and forwarding the packet in a process chain. While conventional pipeline designs provide effective data processing solutions in some applications, packet handling in pipelines may be susceptible to the introduction of unwanted latency and reduction of throughput as packets are passed among pipeline stages.

[0026] Multi-tenant SDN platforms are increasingly common as cloud-computing infrastructure becomes more ubiquitous and cost-effective and new paradigms such as network slicing are utilized. The term “tenant” is a logical separation among entities and may include, for example and not by way of limitation, different organizations, enterprises, customers, or businesses. Different tenants can also be part of the same organization, such as different departments (e.g., engineering, operations, customer service, etc.) in the same company.

[0027] Multi-tenant SDN platforms utilize commonly-shared computing infrastructure and resources that are virtualized to support different discrete tenant domains that are operated separately in accordance with individual application policies. While operations of multi-tenant SDN platforms are typically transparent, tenants nonetheless require high security assurances for network operations and seek isolation for their tenant domain to restrict access to confidential and proprietary information and reduce the size of attack surfaces to bad actors.

[0028] Packet processing pipeline performance is improved with increased security in multi-tenant scenarios by the present principles in which a programmable multi-stage packet processing pipeline executing in a data plane of an SDN network handles data packets according to objects in a directed graph that are programmed using multiple separate namespaces. Each namespace provides isolation for programming in the processing stages to provide protection for the pipeline that may span across multiple different tenant domains. While programming is handled using separate namespaces, the programmed stages are converted into executable code that runs in a single data plane of the SDN network to optimize data packet flow while minimizing latency.

[0029] Turning now to the drawings, FIG. 1 shows an illustrative 5G (5th generation) mobile network **100** that provides an exemplary use case with which the present namespaces for protected multi-tenant packet processing pipelines may be utilized. User equipment (UE) **105**, comprising computing devices, such as phones, tablets, personal computers, laptops, and Internet-of-Things (IoT) devices, communicate wirelessly over an air interface with a radio unit (RU) **110**. The RU delivers data to a radio access network (RAN) **115**. The RAN is commonly implemented virtually on computer networking infrastructure **120** and is referred to as a “vRAN.” For example, the infrastructure may comprise a far edge computing network or platform of a mobile operator (MO) or may be incorporated into on-premises equipment of an enterprise.

[0030] The RAN **115** is connected to a core network (CN) **125** that is implemented, in this illustrative example, in an edge datacenter (DC) **130** using cloud-computing infrastructure. The edge DC is configurable as a multi-access edge computing (MEC) network **135** in some applications, or it may include MEC functions and features. In some implementations, the MEC network may be incorporated in whole or part in the computer networking infrastructure **120** supporting the RAN **115** (not shown). The edge DC is operatively coupled with a central DC **140** using cloud-computing infrastructure that supports a data network (DN) **145** and other computing resources and services.

[0031] FIG. 2 shows an illustrative multi-tenant use case in which different tenants 1, 2, . . . N (as respectively indicated by reference numerals **205**, **210**, and **215**) in commonly shared 5G network infrastructure **120**, **130**, and **140** and associated resources. For example, the RAN **115**, CN **125**, and DN **145** can be sliced on an end-to-end basis and the network slices distributed among the different tenants. However, it is emphasized that network slicing is merely illustrative and that other ways of sharing network infrastructure and resources among tenants may be utilized as needed to meet the needs of a given implementation.

[0032] The 5G network **100** is architected using NFV and SDN principles. FIG. 3 shows an illustrative software-defined networking (SDN) architecture **300** that provides for an application plane **305** that is supported by separate control **310** and data **315** planes (the data plane is also commonly referred to as a user plane in technical literature and applicable standards). The application and control planes interoperate through a northbound API (application programming interface) **320** and the control and data planes interoperate through a southbound API **325**.

[0033] The application plane **305** supports sets of various applications **330**, **335**, and **340** that are respectively associ-

ated with the tenants **205**, **210**, and **215**. The types, features, and functions of the applications can vary by implementation. Exemplary applications deal with security **345**, routing **350**, traffic engineering **355**, quality of service (QoS) **360**, multiprotocol label switching **365**, policies **370**, and other suitable subjects **375**. The tenant applications are executed separately and in isolation from each other such that dedicated network resources are not shared between programs, and the execution of one tenant program does not impact execution of others. For security purposes, for example, access by packets controlled by one program to data stored by another is typically limited.

[0034] The data plane **315** comprises data-forwarding network elements **385** of one or more portions of computing infrastructure **120**, **130**, and **140** that underlies the SDN architecture **300**. Examples of such network elements include, for example, switches, routers, firewalls, gateways, and the like. The data plane is responsible for actual movement of data from one system to another and forwards data based on information provided by an SDN controller **380** in the control plane **310**.

[0035] The data plane **315** performs a series of operations, including the parsing of packet headers, determining a sequence of processing operations to be applied, and forwarding the packet based on the results of the operations. FIG. 4 shows a taxonomy **400** of illustrative basic data plane functions for packet processing including, for example and without limitation, parsing **405**, classification **410**, modification **415**, deparsing **420**, and forwarding **425**. Additional data plane functions can support other services such as scheduling, filtering, metering, and traffic shaping.

[0036] Parsing is a process of locating protocol headers in the packet buffer and extracting the relevant header fields into packet descriptors (i.e., metadata). Field values are used in classification to match the packet with the corresponding forwarding policy, which describes the forwarding or processing actions to be applied to the packet (e.g., which output interfaces to use, whether to drop the packet, etc.) and the required packet modification actions (e.g., rewriting a header field).

[0037] Modification applies the actions retrieved during classification, and can update an internal state, for example, to increment a flow counter. After modifications are applied, packet headers may be re-generated from packet descriptors in deparsing. The packet is sent to a suitable output interface for transmission during forwarding. Forwarding may include application of scheduling policies, for example to enforce network-level QoS policies, and traffic shaping to limit the amount of network resources a given flow or tenant application may consume. The combination of classification, modification, and forwarding based on match-action rules **435** is commonly referred to as match-action processing **430**.

[0038] FIG. 5 shows an illustrative example of multi-tenant service function chaining using the SDN controller **380** which is configured to support user programming **505**. Service function chaining leverages NFV and SDN to create a service overlay on top of the physical network elements in the data plane **315** as described, for example, by the IETF (Internet Engineering Task Force) RFC 7665. In service function chaining, data packet flows through the data plane are steered between service functions (SF) that are instantiated at different places in the network. Pipes are built between SF forwarding elements in the overlay using, for

example, IP (Internet Protocol) tunnels, MPLS label-switched paths or pseudowires. SFC (service function chaining) encapsulation is provided using a network service header to indicate to which chain of service functions the packet has been assigned and which service function is utilized next.

[0039] As shown in FIG. 5, an SFC-enabled data plane **510** supports different services 1 and 2 for respective different tenants 1 and 2. The flow path **515** for service 1 chains together various SFs (representatively indicated by reference numeral **520**) through which data packets **525** for tenant 1 are processed. The flow path **530** for service 2 chains together other SFs through which data packets **535** for tenant 2 are processed. In conventional service function chaining, the flow paths are set in the data plane **510** according to data delivery rules **540** provided by the SDN controller **380** over the southbound API **325**.

[0040] FIG. 6 shows a programmable data plane **605** in which a multi-tenant protected packet processing pipeline **610** is operated in accordance with a directed graph **615**. In comparison to conventional service function chaining as shown in FIG. 5 and described above, a user programming interface **620** is not supported in the control plane **310**. Instead, user programming capabilities extend to the packet processing pipeline to create and configure graph objects **625** in the directed graph. The graph objects include nodes (representatively indicated by reference numeral **630**) and edges **635**. The nodes represent match-action classifiers and the edges are matches that represent packet flows. The packet processing pipeline receives a stream of tenant packets **525** and **535** at the input **640** to the pipeline, processes the packets in accordance with the directed graph, and provides a stream of tenant packets at the output **645** of the pipeline.

[0041] FIG. 7 shows an illustrative directed graph **700** for implementing a service using the packet processing pipeline **610**. The directed graph uses a global namespace which is programmed through a single user programming domain **705**. The directed graph includes graph objects **710** comprising illustrative nodes **715**, **720**, and **725** and edges **730**, **735**, and **740**. Node **715** is a GTP (GPRS tunneling protocol) stage that functions to add new headers to the front of a packet. Node **720** is a routing stage that performs longest prefix match and chooses a suitable output interface (a longest prefix match is a routing algorithm that determines the most specific route for a data packet). Node **725** is an IPsec (Internet Protocol Security) stage that encrypts packets using keys stored in the programming.

[0042] The graph objects **710** are programmable through respective programmers in the user programming domain **705**. These include a session programmer **745** that programs the session configuration, for example, GTP tunnel properties. A routing programmer **750** programs routing configuration for the routing stage **720**. An IPsec programmer **755** programs keys for different IPsec tunnels.

[0043] The global namespace and single user programming domain can provide satisfactory results in some scenarios, but is less than optimal for use cases that utilize the zero-trust model in which security is implemented by assigning the least required access to perform specific tasks. In this illustrative example, the graph objects **710** in the global graph for service programming are accessible by each of the programmers **745**, **750**, and **755**. Thus, for example, the component adding the GTP headers would also have

access to the IPsec component of the programming including the keys used for encryption.

[0044] Rather than use a single global graph and associated namespace, the present packet processing pipeline principles are based on the utilization of multiple separate and individual namespaces. FIG. 8 shows an illustrative namespace **805** that encapsulates graph objects **810** within a predefined scope in which programming for the objects is bound to the namespace. The namespace defines one or more public APIs **815** to interoperate with programming in other namespaces (representatively indicated by reference numeral **820**) and further defines one or more public interfaces **825** to SDN resources **830** and/or services.

[0045] Programming of the graph objects (i.e., nodes and edges) can be linked from different namespaces. For example, as shown in FIGS. 9A and 9B, a given edge **905** is linked to two nodes **910** and **915** in a directed graph **920**. FIG. 9A shows an illustrative example of a first type of programming link in which the edge **905** is programmed in a namespace B **930** that is different from the namespace A **925** of node **910** from which the edge originates. FIG. 9B shows an illustrative example of a second type of programming link in which the edge is programmed in namespace A that is different from the namespace B for node **915** to which the edge leads.

[0046] FIG. 10 shows an illustrative directed graph **1000** for implementing a service using the packet processing pipeline **610**. The directed graph is programmed through separate and individual user programming domains **1005**, **1010**, and **1015** using multiple namespaces. These include a session namespace **1020**, a routing namespace **1025**, and an IPsec namespace **1030**. The directed graph includes graph objects **1035** comprising illustrative nodes **1040**, **1045**, and **1050** and edges **1055**, **1060**, and **1065**. The nodes provide respective processing stages in the pipeline including GTP encapsulation, routing, and IPsec encryption.

[0047] The graph objects **1035** are programmable through respective programmers **1070**, **1075**, and **1080** in the respective individual user programming domains **1005**, **1010**, and **1015**. The programmers operate on the graph objects through respective separate and individual user programming interfaces **1085**, **1090**, and **1095**. The provisioning of the separate namespaces and associated programmers and interfaces ensures that isolation is enforced among groups of graph objects, which are only programmable by users (and/or tenants) through assigned programming interfaces to the associated namespaces.

[0048] The binding of programming to namespaces and the isolation provided by the separate programming domains provide protection to the packet processing pipeline to address the security issues arising with the single global namespace. Different users and/or tenants are restricted to programming of only parts of the pipeline to which they are authenticated and authorized to access. The protected packet processing pipeline adheres to the zero-trust model because if any one part of the directed graph programming is compromised or altered (e.g., by a malicious attack), the negative impact is contained to just the affected part and cannot spread to other programming.

[0049] The use of multiple namespaces and associated programming domains may be extended to create more complex services. FIG. 11 shows an illustrative directed graph **1100** that is composed of multiple constituent services that are programmed using a layered approach in which

some namespaces are hidden from some users. In this example, the graph uses the routing namespace **1025** and IPsec namespace **1030** that are respectively associated with the routing programmer **1075** and IPsec programmer **1080**, as in the previous example.

[0050] There are two additional services, A and B, that are utilized, as shown in the drawing, in which service A uses a single namespace **1105** that contains service A graph objects **1110** which are accessed via a service A graph object programmer **1115**. The namespace **1120** for service B contains two namespaces within it—a service B1 namespace **1125** and a service B2 namespace **1130** that are respectively associated with a service B1 graph object programmer **1135** and a service B2 graph object programmer **1140**. The service B1 namespace contains service B1 graph objects **1145** and the service B2 namespace contains service B2 graph objects **1150**. The graph objects for services B1 and B2 are combined into service B graph objects **1155**. As discussed above, the service programming for each namespace is mapped into the single directed graph **1100** that controls processing of the packet processing pipeline **610**.

[0051] The packet processing pipeline **610** supports flows through either service A or B programming based on application classification **1160**. The traffic flows from the application classification to a given service based on the graph objects contained in that service.

[0052] The illustrative example shown in FIG. 11 highlights how services can be built up from programming in different namespaces that can be layered or nested. In addition, some namespaces can be hidden or private. For example, the service B graph objects use objects from the B1 and B2 namespaces that are not visible, so these namespaces can be considered private to service B.

[0053] FIG. 12 is a flowchart of an illustrative method **1200** for operating a data packet processing pipeline in a network. Unless specifically stated, methods or steps shown in the flowchart blocks and described in the accompanying text are not constrained to a particular order or sequence. In addition, some of the methods or steps thereof can occur or be performed concurrently and not all the methods or steps have to be performed in a given implementation depending on the requirements of such implementation and some methods or steps may be optionally utilized.

[0054] Block **1205** includes instantiating the data packet processing pipeline on a programmable data plane of the computing device, the pipeline having a plurality of processing stages comprising configurable graph objects arranged in a directed graph, wherein the pipeline processes data packets through the plurality of processing stages according to the directed graph. Block **1210** includes enabling programming of the graph objects in the processing stages using a plurality of unique namespaces, each namespace encapsulating associated graph object programming within a predefined scope. Block **1215** includes providing a plurality of user programming interfaces to the directed graph, each user programming interface being associated with a namespace corresponding with one or more graph objects, wherein programming of graph objects through a user programming interface is bound to its associated namespace to provide programming isolation among the processing stages.

[0055] FIG. 13 is a flowchart of an illustrative method **1300** that may be executed by a computing device in an SDN network. Block **1305** includes implementing a multi-stage

data packet processing pipeline that is supported across a plurality of domains respectively associated with a plurality of different tenants. Block **1310** includes utilizing a directed graph of configurable graph objects to control operations of stages of the data packet processing pipeline, the graph objects including nodes defining match-action classifiers and edges defining data packet flows through the data packet processing pipeline based on matches, and wherein sets of one or more graph objects are grouped within respective discrete namespaces in the directed graph.

[0056] Block **1315** includes associating a unique user programming interface, among a plurality of unique user programming interfaces, with each set of graph objects as grouped by discrete namespace, wherein each unique user programming interface is limited to programming for only an associated set of graph objects. Block **1320** includes assigning ones of the plurality of unique user programming interfaces to respective ones of the different tenants, each tenant being enabled to program stages of the data packet processing pipeline using an assigned unique user programming interface.

[0057] FIG. **14** is a flowchart of an illustrative method **1400** that may be performed by a computing device. Block **1405** includes implementing a data packet processing pipeline supported in a cloud-computing network configured with an SDN architecture supporting multiple tenants, wherein the data packet processing pipeline is programmatically-operated according to a directed graph. Block **1410** includes supporting a user programming domain for the data packet processing pipeline in which the directed graph comprises user-programmed graph objects grouped into programming containers identified by unique namespaces.

[0058] Block **1415** includes providing access to tenants to selected ones of the namespaces to generate programming for the data packet processing pipeline with user-programmed graph objects. Block **1420** includes supporting an execution domain for the data packet processing pipeline in which the programming is converted into executable code running on a single data plane of physical network infrastructure in the cloud-computing network.

[0059] FIG. **15** shows an illustrative architecture **1500** for a computing device, such

[0060] as a server, capable of executing the various components described herein for the present namespaces for protected multi-tenant packet processing pipelines. The architecture **1500** illustrated in FIG. **15** includes one or more processors **1502** (e.g., central processing unit, dedicated AI chip, graphics processing unit, etc.), a system memory **1504**, including RAM (random access memory) **1506** and ROM (read only memory) **1508**, and a system bus **1510** that operatively and functionally couples the components in the architecture **1500**. A basic input/output system containing the basic routines that help to transfer information between elements within the architecture **1500**, such as during startup, is typically stored in the ROM **1508**. The architecture **1500** further includes a mass storage device **1512** for storing software code or other computer-executed code that is utilized to implement applications, a file system, and an operating system (OS). The mass storage device **1512** is connected to the processor **1502** through a mass storage controller (not shown) connected to the bus **1510**. The mass storage device **1512** and its associated computer-readable storage media provide non-volatile storage for the architecture **1500**. Although the description of computer-readable

storage media contained herein refers to a mass storage device, such as a hard disk or CD-ROM drive, it may be appreciated by those skilled in the art that computer-readable storage media can be any available storage media that can be accessed by the architecture **1500**.

[0061] By way of example, and not limitation, computer-readable storage media may include volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, or other data. For example, computer-readable media includes, but is not limited to, RAM, ROM, EPROM (erasable programmable read only memory), EEPROM (electrically erasable programmable read only memory), Flash memory or other solid state memory technology, CD-ROM, DVDs, HD-DVD (High Definition DVD), Blu-ray or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by the architecture **1500**.

[0062] According to various embodiments, the architecture **1500** may operate in a networked environment using logical connections to remote computers through a network. The architecture **1500** may connect to the network through a network interface unit **1516** connected to the bus **1510**. It may be appreciated that the network interface unit **1516** also may be utilized to connect to other types of networks and remote computer systems. The architecture **1500** also may include an input/output controller **1518** for receiving and processing input from a number of other devices, including a keyboard, mouse, touchpad, touchscreen, control devices such as buttons and switches or electronic stylus (not shown in FIG. **15**). Similarly, the input/output controller **1518** may provide output to a display screen, user interface, a printer, or other type of output device (also not shown in FIG. **15**).

[0063] It may be appreciated that the software components described herein may, when loaded into the processor **1502** and executed, transform the processor **1502** and the overall architecture **1500** from a general-purpose computing system into a special-purpose computing system customized to facilitate the functionality presented herein. The processor **1502** may be constructed from any number of transistors or other discrete circuit elements, which may individually or collectively assume any number of states. More specifically, the processor **1502** may operate as a finite-state machine, in response to executable instructions contained within the software modules disclosed herein. These computer-executable instructions may transform the processor **1502** by specifying how the processor **1502** transitions between states, thereby transforming the transistors or other discrete hardware elements constituting the processor **1502**.

[0064] Encoding the software modules presented herein also may transform the physical structure of the computer-readable storage media presented herein. The specific transformation of physical structure may depend on various factors, in different implementations of this description. Examples of such factors may include, but are not limited to, the technology used to implement the computer-readable storage media, whether the computer-readable storage media is characterized as primary or secondary storage, and the like. For example, if the computer-readable storage media is implemented as semiconductor-based memory, the software disclosed herein may be encoded on the computer-

readable storage media by transforming the physical state of the semiconductor memory. For example, the software may transform the state of transistors, capacitors, or other discrete circuit elements constituting the semiconductor memory. The software also may transform the physical state of such components in order to store data thereupon.

[0065] As another example, the computer-readable storage media disclosed herein may be implemented using magnetic or optical technology. In such implementations, the software presented herein may transform the physical state of magnetic or optical media, when the software is encoded therein. These transformations may include altering the magnetic characteristics of particular locations within given magnetic media. These transformations also may include altering the physical features or characteristics of particular locations within given optical media to change the optical characteristics of those locations. Other transformations of physical media are possible without departing from the scope and spirit of the present description, with the foregoing examples provided only to facilitate this discussion.

[0066] In light of the above, it may be appreciated that many types of physical transformations take place in the architecture 1500 in order to store and execute the software components presented herein. It also may be appreciated that the architecture 1500 may include other types of computing devices, including wearable devices, handheld computers, embedded computer systems, smartphones, PDAs, and other types of computing devices known to those skilled in the art. It is also contemplated that the architecture 1500 may not include all of the components shown in FIG. 15, may include other components that are not explicitly shown in FIG. 15, or may utilize an architecture completely different from that shown in FIG. 15.

[0067] FIG. 16 is a high-level block diagram of an illustrative datacenter 1600 that provides cloud computing services or distributed computing services that may be used to implement the present namespaces for protected multi-tenant packet processing pipelines. Datacenter 1600 may incorporate one or more of the features disclosed in the DCs shown in the drawings and described in the accompanying text. A plurality of servers 1601 are managed by datacenter management controller 1602. Load balancer 1603 distributes requests and computing workloads over servers 1601 to avoid a situation wherein a single server may become overwhelmed. Load balancer 1603 maximizes available capacity and performance of the resources in datacenter 1600. Routers/switches 1604 support data traffic between servers 1601 and between datacenter 1600 and external resources and users (not shown) via an external network 1605, which may be, for example, a local area network (LAN) or the Internet.

[0068] Servers 1601 may be standalone computing devices, and/or they may be configured as individual blades in a rack of one or more server devices. Servers 1601 have an input/output (I/O) connector 1606 that manages communication with other database entities. One or more host processors 1607 on each server 1601 run a host OS 1608 that supports multiple virtual machines (VM) 1609. Each VM 1609 may run its own OS so that each VM OS 1610 on a server is different, or the same, or a mix of both. The VM OSs 1610 may be, for example, different versions of the same OS (e.g., different VMs running different current and legacy versions of the Windows® operating system). In addition, or alternatively, the VM OSs 1610 may be pro-

vided by different manufacturers (e.g., some VMs running the Windows® operating system, while other VMs are running the Linux® operating system). Each VM 1609 may also run one or more applications (Apps) 1611. Each server 1601 also includes storage 1612 (e.g., hard disk drives (HDD)) and memory 1613 (e.g., RAM) that can be accessed and used by the host processors 1607 and VMs 1609 for storing software code, data, etc.

[0069] Datacenter 1600 provides pooled resources on which customers or tenants can dynamically provision and scale applications as needed without having to add servers or additional networking. This allows tenants to obtain the computing resources they need without having to procure, provision, and manage infrastructure on a per-application, ad-hoc basis. A cloud computing datacenter 1600 allows tenants to scale resources up or down dynamically to meet the current needs of their businesses. Additionally, a datacenter operator can provide usage-based services to tenants so that they pay for only the resources they use, when they need to use them. For example, a tenant may initially use one VM 1609 on server 1601₁ to run their applications 1611. When demand for an application 1611 increases, the datacenter 1600 may activate additional VMs 1609 on the same server 1601₁ and/or on a new server 1601_N as needed. These additional VMs 1609 can be deactivated if demand for the application later drops.

[0070] Datacenter 1600 may offer guaranteed availability, disaster recovery, and back-up services. For example, the datacenter may designate one VM 1609 on server 1601₁ as the primary location for the tenant's application and may activate a second VM 1609 on the same or a different server as a standby or back-up in case the first VM or server 1601₁ fails. The datacenter management controller 1602 automatically shifts incoming user requests from the primary VM to the back-up VM without requiring tenant intervention. Although datacenter 1600 is illustrated as a single location, it will be understood that servers 1601 may be distributed to multiple locations across the globe to provide additional redundancy and disaster recovery capabilities. Additionally, datacenter 1600 may be an on-premises, private system that provides services to a single enterprise user or may be a publicly accessible, distributed system that provides services to multiple, unrelated customers and tenants or may be a combination of both.

[0071] Domain Name System (DNS) server 1614 resolves domain and host names into IP addresses for all roles, applications, and services in datacenter 1600. DNS log 1615 maintains a record of which domain names have been resolved by role. It will be understood that DNS is used herein as an example and that other name resolution services and domain name logging services may be used to identify dependencies, for example, in other embodiments, IP or packet sniffing, code instrumentation, or code tracing.

[0072] Datacenter health monitoring 1616 monitors the health of the physical systems, software, and environment in datacenter 1600. Health monitoring 1616 provides feedback to datacenter managers when problems are detected with servers, blades, processors, or applications in datacenter 1600, or when network bandwidth or communications issues arise.

[0073] Access control service 1617 determines whether users are allowed to access particular connections and services provided at the datacenter 1600. Directory and

identity management service **1618** authenticates user credentials for tenants on datacenter **1600**.

[0074] FIG. 17 shows an illustrative mobile network that uses a service-based architecture (SBA) **1700** as defined by the 3GPP (3rd Generation Partnership Project). An SBA provides a modular framework from which common applications can be deployed using components of varying sources and vendors. Control plane functionality and common data repositories of the network are delivered by way of a set of Network Functions (NFs) that are interconnected with a service-based interface bus, in which each has authorization to access each other's services. Assuming the role of either service consumer or service producer, NFs are self-contained, independent, and reusable. Each NF service exposes its functionality through a Service Based Interface (SBI), which employs a well-defined REST (Representational State Transfer) interface using HTTP/2 (Hypertext Transfer Protocol Version 2).

[0075] As shown, the mobile network includes a CN **1702** that is interoperable with 4G (4th generation) and 5G RANs **1704** and **1706** that support wireless communications with UE **1701**. The SBA architecture **1700** supports a 5G next generation core (NGC) network that includes 4G evolved packet core (EPC) instances to enable some 4G LTE (Long Term Evolution) use cases when implementing the present principles. With 4G mode, some 5G components such as the UDR (unified data repository) **1705** and UPF (user plane function) **1710** support 4G mode without the need to revert to a legacy 4G stack. Other 4G components include an MME (mobility management entity) **1780** and IWF (interworking function) **1785**.

[0076] The UPF handles user data, performing operations such as maintaining PDU (Protocol Data Unit) sessions, packet routing and forwarding, packet inspection, policy enforcement for the user plane, QoS handling, traffic usage reporting for billing, and the like. The UPF further provides an interconnection point between the mobile network infrastructure and an external DN **1765**. The AF (application function) **1770** provides service or application related information to a VNF service consumer, for example, a mobile network or enterprise operator **1775**.

[0077] The AMF (access and mobility management function) **1715** receives all connection and session related information from the UE **1701** but is only responsible for handling connection and mobility management tasks such as registration and authentication, identification, and mobility. All messages related to session management are forwarded over an interface to the Session Management Function (SMF) **1720** that establishes and manages sessions. It also selects and controls the UPF **1710** and handles paging. The AF **1770** provides service or application related information to the NF service consumer. For example, the AF performs operations such as retrieving resources and exposing services to end-users. Other 3GPP-defined 5G network functions in the architecture **1700** include SMF (session management function) **1720**; PCF (policy control function) **1725**; AUSF (authentication server function) **1730**; UDM (unified data management) **1735**; and NRF (network repository function) **1740**.

[0078] Various exemplary embodiments of the present namespaces for protected multi-tenant packet processing pipelines are now presented by way of illustration and not as an exhaustive list of all embodiments. An example includes a method, operable on a computing device, for operating a

data packet processing pipeline in a network, comprising: instantiating the data packet processing pipeline on a programmable data plane of the computing device, the pipeline having a plurality of processing stages comprising configurable graph objects arranged in a directed graph, wherein the pipeline processes data packets through the plurality of processing stages according to the directed graph; enabling programming of the graph objects in the processing stages using a plurality of unique namespaces, each namespace encapsulating associated graph object programming within a predefined scope; and providing a plurality of user programming interfaces to the directed graph, each user programming interface being associated with a namespace corresponding with one or more graph objects, wherein programming of graph objects through a user programming interface is bound to its associated namespace to provide programming isolation among the processing stages.

[0079] In another example, the method further includes distributing the plurality of programming interfaces to the directed graph across different tenant domains. In another example, the graph objects comprise nodes, including match-action classifiers and interfaces, and further comprise edges including matches that represent data flows through the processing stages. In another example, the match-action classifiers apply match-action rules to process data packets. In another example, the method further includes converting the directed graph into code that executes on the programmable data plane to operate the data packet processing pipeline. In another example, each namespace is further arranged to define an application programming interface (API) that is globally accessible by other namespaces and further defining interfaces to resources in the network. In another example, the computing device is incorporated into infrastructure underlying a software-defined networking (SDN) network.

[0080] A further example includes one or more hardware-based non-transitory computer-readable memory devices storing computer-executable instructions which, upon execution by one or more processors disposed in a computing device in a software-defined networking (SDN) network, cause the computing device to: implement a multi-stage data packet processing pipeline that is supported across a plurality of domains respectively associated with a plurality of different tenants; utilize a directed graph of configurable graph objects to control operations of stages of the data packet processing pipeline, the graph objects including nodes defining match-action classifiers and edges defining data packet flows through the data packet processing pipeline based on matches, and wherein sets of one or more graph objects are grouped within respective discrete namespaces in the directed graph; associate a unique user programming interface, among a plurality of unique user programming interfaces, with each set of graph objects as grouped by discrete namespace, wherein each unique user programming interface is limited to programming for only an associated set of graph objects; and assign ones of the plurality of unique user programming interfaces to respective ones of the different tenants, each tenant being enabled to program stages of the data packet processing pipeline using an assigned unique user programming interface.

[0081] In another example, each discrete namespace provides encapsulation for the programming of the associated set of graph objects to restrict access from other ones of the plurality of the unique programming interfaces. In another

example, the one or more hardware-based non-transitory computer-readable memory devices further include configuring one or more of the discrete namespaces to define public application programming interfaces (APIs). In another example, the one or more hardware-based non-transitory computer-readable memory devices further include configuring one or more of the discrete namespaces to define public interfaces to SDN network resources. In another example, the match-action classifiers specify filtering criteria for data packets and actions for performing responsive to a data packet matching specified filtering criteria. In another example, the match-action classifiers apply match-action rules to process data packets in the data packet processing pipeline. In another example, the match-action rules comprise one of exact match, longest-prefix match, bloom filter match, or regular expression match. In another example, the data packet processing pipeline is instantiated in a programmable data plane in the SDN network.

[0082] A further example includes a computing device, comprising: a processor; a hardware-based non-transitory computer-readable storage device having computer-executable instructions stored thereon which, when executed by the processor, cause the computing device to: implement a data packet processing pipeline supported in a cloud-computing network configured with an SDN (software-defined networking) architecture supporting multiple tenants, wherein the data packet processing pipeline is programmatically-operated according to a directed graph; support a user programming domain for the data packet processing pipeline in which the directed graph comprises user-programmed graph objects grouped into programming containers identified by unique namespaces; provide access to tenants to selected ones of the namespaces to generate programming for the data packet processing pipeline with user-programmed graph objects; and support an execution domain for the data packet processing pipeline in which the programming is converted into executable code running on a single data plane of physical network infrastructure in the cloud-computing network.

[0083] In another example, the graph objects include nodes comprising processing elements and edges comprising packet flow among the processing elements. In another example, the namespaces define public interfaces accessible by programming in the namespaces and interfaces to SDN resources. In another example, the computing device is instantiated in physical network infrastructure used in a 4G (4th generation) or a 5G (5th generation) mobile network. In another example, the physical network infrastructure comprises packet-forwarding devices including one of switch, router, or firewall.

[0084] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

What is claimed:

1. A method, operable on a computing device, for operating a data packet processing pipeline in a network, comprising:

instantiating the data packet processing pipeline on a programmable data plane of the computing device, the

pipeline having a plurality of processing stages comprising configurable graph objects arranged in a directed graph, wherein the pipeline processes data packets through the plurality of processing stages according to the directed graph;

enabling programming of the graph objects in the processing stages using a plurality of unique namespaces, each namespace encapsulating associated graph object programming within a predefined scope; and

providing a plurality of user programming interfaces to the directed graph, each user programming interface being associated with a namespace corresponding with one or more graph objects, wherein programming of graph objects through a user programming interface is bound to its associated namespace to provide programming isolation among the processing stages.

2. The method of claim 1 further including distributing the plurality of programming interfaces to the directed graph across different tenant domains.

3. The method of claim 1 in which the graph objects comprise nodes, including match-action classifiers and interfaces, and further comprise edges including matches that represent data flows through the processing stages.

4. The method of claim 3 in which the match-action classifiers apply match-action rules to process data packets.

5. The method of claim 1 further including converting the directed graph into code that executes on the programmable data plane to operate the data packet processing pipeline.

6. The method of claim 1 further arranging each namespace to define an application programming interface (API) that is globally accessible by other namespaces and further defining interfaces to resources in the network.

7. The method of claim 1 in which the computing device is incorporated into infrastructure underlying a software-defined networking (SDN) network.

8. One or more hardware-based non-transitory computer-readable memory devices storing computer-executable instructions which, upon execution by one or more processors disposed in a computing device in a software-defined networking (SDN) network, cause the computing device to:

implement a multi-stage data packet processing pipeline that is supported across a plurality of domains respectively associated with a plurality of different tenants;

utilize a directed graph of configurable graph objects to control operations of stages of the data packet processing pipeline, the graph objects including nodes defining match-action classifiers and edges defining data packet flows through the data packet processing pipeline based on matches, and wherein sets of one or more graph objects are grouped within respective discrete namespaces in the directed graph;

associate a unique user programming interface, among a plurality of unique user programming interfaces, with each set of graph objects as grouped by discrete namespace, wherein each unique user programming interface is limited to programming for only an associated set of graph objects; and

assign ones of the plurality of unique user programming interfaces to respective ones of the different tenants, each tenant being enabled to program stages of the data packet processing pipeline using an assigned unique user programming interface.

9. The one or more hardware-based non-transitory computer-readable memory devices of claim 8 in which each

discrete namespace provides encapsulation for the programming of the associated set of graph objects to restrict access from other ones of the plurality of the unique programming interfaces.

10. The one or more hardware-based non-transitory computer-readable memory devices of claim **8** further including configuring one or more of the discrete namespaces to define public application programming interfaces (APIs).

11. The one or more hardware-based non-transitory computer-readable memory devices of claim **8** further including configuring one or more of the discrete namespaces to define public interfaces to SDN network resources.

12. The one or more hardware-based non-transitory computer-readable memory devices of claim **8** in which the match-action classifiers specify filtering criteria for data packets and actions for performing responsive to a data packet matching specified filtering criteria.

13. The one or more hardware-based non-transitory computer-readable memory devices of claim **8** in which the match-action classifiers apply match-action rules to process data packets in the data packet processing pipeline.

14. The one or more hardware-based non-transitory computer-readable memory devices of claim **13** wherein the match-action rules comprise one of exact match, longest-prefix match, bloom filter match, or regular expression match.

15. The one or more hardware-based non-transitory computer-readable memory devices of claim **8** in which the data packet processing pipeline is instantiated in a programmable data plane in the SDN network.

16. A computing device, comprising:

a processor;

a hardware-based non-transitory computer-readable storage device having computer-executable instructions stored thereon which, when executed by the processor, cause the computing device to:

implement a data packet processing pipeline supported in a cloud-computing network configured with an SDN (software-defined networking) architecture supporting multiple tenants, wherein the data packet processing pipeline is programmatically-operated according to a directed graph;

support a user programming domain for the data packet processing pipeline in which the directed graph comprises user-programmed graph objects grouped into programming containers identified by unique namespaces;

provide access to tenants to selected ones of the namespaces to generate programming for the data packet processing pipeline with user-programmed graph objects; and

support an execution domain for the data packet processing pipeline in which the programming is converted into executable code running on a single data plane of physical network infrastructure in the cloud-computing network.

17. The computing device of claim **16** in which the graph objects include nodes comprising processing elements and edges comprising packet flow among the processing elements.

18. The computing device of claim **17** in which the namespaces define public interfaces accessible by programming in the namespaces and interfaces to SDN resources.

19. The computing device of claim **16** instantiated in physical network infrastructure used in a 4G (4th generation) or a 5G (5th generation) mobile network.

20. The computing device of claim **19** in which the physical network infrastructure comprises packet-forwarding devices including one of switch, router, or firewall.

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